

CHOP Workshop

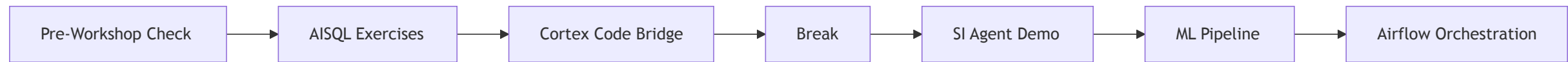
Snowflake AI · Pharmacy Intelligence · ML Pipeline

Audience: CHOP Data Analysts + Data Scientists

Format: 2 hours · Hands-on exercises + Live demos

SE Note: Open `CHOP_workshop_plan.html` for the full reference guide. This deck is the running-order script.

Workshop Flow — 2 Hours



Time	Block	Format	File
0:00–0:05	Env check (Cell 1)	Everyone runs notebook Cell 1	<code>workshop_aisql_si_exercises.ipynb</code>
0:05–0:30	SE walkthrough — AISQL concepts	SE presents	This deck
0:30–1:00	Hands-on AISQL exercises (Cells 2–4)	Participants run notebook	<code>workshop_aisql_si_exercises.ipynb</code>
0:55–1:00	Cortex Code bridge (Cell 5)	SE uses coco CLI live	This deck
1:00–1:10	Break	—	—
1:10–1:40	SI Agent demo (Cell 6)	SE drives, participants ask	<code>workshop_aisql_si_exercises.ipynb</code>
1:40–2:00	ML Pipeline + Airflow	SE demo, optional hands-on	This deck

Pre-Workshop Setup

Run 1 week before. Participants self-verify before they arrive.

Pre-Workshop: Participant Readiness Check

SE Action: Send `sql/CHOP_participant_readiness_check.sql` to all participants. Ask for a screenshot by EOD the day before.

Section A — Data Analysts (`CHOP_SNOW_INTELLIGENCE`)

Check	What it verifies
A-1	Role is active
A-2	Warehouse active
A-3	<code>V_PHARMACY_ALLMEDICALSCRIPTS</code> has rows
A-4	<code>V_PHARMACYDRUG_MASTER</code> has rows
A-5	<code>AI_CLASSIFY</code> callable (<i>credits: ~\$0.001</i>)
A-6	<code>EXTRACT_PRESCRIPTION_ENTITIES</code> UDF callable
A-7	Snowsight → Agents → confirm agent visible

If any check FAILS: Contact admin before workshop day. See admin-routine memory for script order (00–12).

Section B — Data Scientists (`ML_ENGINEER`)

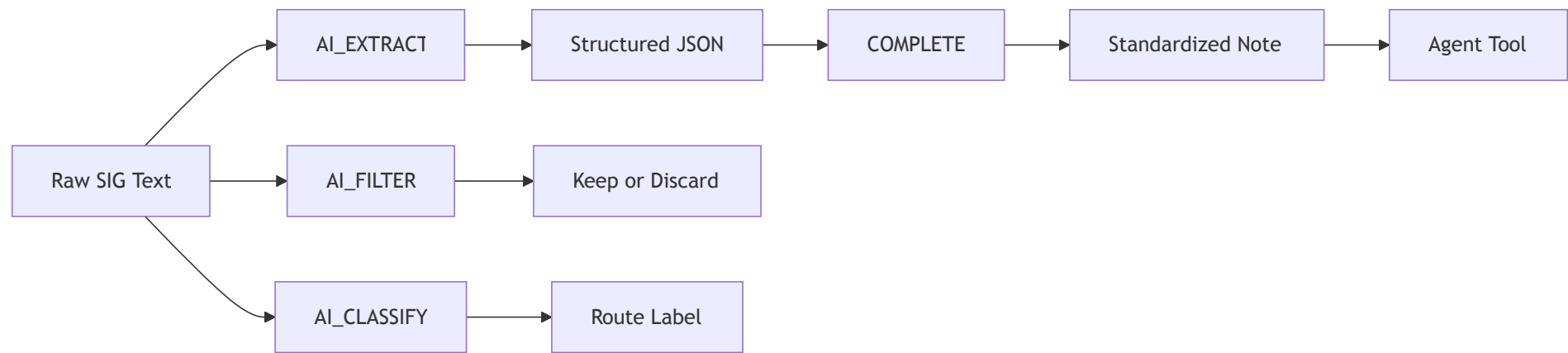
Check	What it verifies
B-1	Role is active
B-2	Warehouse active
B-3	<code>ADMISSIONS</code> table has rows
B-4	<code>PATIENTS</code> table has rows
B-5	<code>AI_CLASSIFY</code> callable
B-6	Snowsight → Agents → confirm agent visible

AISQL Block

Hands-on Exercises — Cells 2–4 · ~30 min

Both roles participate. Analysts use pharmacy views; Scientists use ML tables.

AISQL: Four Functions, One Flow



Function	What it does	Returns	Analyst use	Scientist use
AI_EXTRACT	Pull structured fields from free text	JSON OBJECT	Extract entities from SIG text	Build narrative from columns → extract
AI_CLASSIFY	Assign text to a controlled label list	JSON OBJECT (:label)	Classify route vs stored ADMINROUTE	Classify discharge → readmission risk tier
AI_FILTER	Boolean quality gate	TRUE / FALSE	Filter incomplete SIG records	Filter low-quality clinical notes
COMPLETE	Free-form LLM transformation	VARCHAR	Standardize informal SIG	Summarize clinical data

SE Note: All 4 functions require only `CORTEX_USER` database role — already granted to both `CHOP_SNOW_INTELLIGENCE` and `ML_ENGINEER`.

Exercise 1: AI_EXTRACT — Notebook Cell 2 (~10 min)

SE: Walk through Part A together first (inline text — both roles). Then let participants run Part B for their role.

Part A — Both roles run this (inline, no table access needed):

```
SELECT SNOWFLAKE.CORTEX.AI_EXTRACT(
  'Administer methotrexate 25mg subcutaneously once weekly.
  Monitor LFTs monthly. Hold if WBC < 3.0.',
  ['medication_name', 'dose', 'dose_unit', 'route',
   'frequency', 'monitoring_instructions', 'hold_condition']
) AS extracted_entities;
```

Part B — Analysts (CHOP_SNOW_INTELLIGENCE):

```
SELECT ADMINISTRATIONDIRECTIONS AS raw_sig,
       SNOWFLAKE.CORTEX.AI_EXTRACT(
         ADMINISTRATIONDIRECTIONS,
         ['medication_name', 'dose', 'route',
          'frequency', 'duration']
       )::VARCHAR AS structured_entities
FROM SI_CHOP.CHOP_SNOW_INTELLIGENCE
     .V_PHARMACY_ALLMEDICALSCRIPTS
WHERE ADMINISTRATIONDIRECTIONS IS NOT NULL
LIMIT 3;
```

Part B — Scientists (ML_ENGINEER):

```
SELECT
  'Patient admitted with '||PRIMARY_DIAGNOSIS||
  ', discharged to '||DISCHARGE_DISPOSITION||
  ', LOS '||LENGTH_OF_STAY||' days.' AS narrative,
  SNOWFLAKE.CORTEX.AI_EXTRACT(
    'Patient admitted with '||PRIMARY_DIAGNOSIS||
    ', discharged to '||DISCHARGE_DISPOSITION||
    ', LOS '||LENGTH_OF_STAY||' days.',
    ['primary_condition', 'discharge_setting',
     'readmission_risk_factors']
  )::VARCHAR AS extracted
FROM HEALTHCARE_ML.RAW_DATA.ADMISSIONS LIMIT 3;
```

Talking point: The EXTRACT_PRESCRIPTION_ENTITIES UDF in the SI agent wraps this exact call. Analysts trigger it by asking a question — no SQL needed.

Exercise 2: AI_CLASSIFY — Notebook Cell 3 (~8 min)

SE: Show Part A, discuss the guardrail concept, then have participants run Part B for their role.

Part A — Both roles (inline):

```
SELECT SNOWFLAKE.CORTEX.AI_CLASSIFY(  
    'infuse 500ml normal saline over 4 hours through peripheral IV line',  
    ['oral','intravenous','intramuscular','subcutaneous','topical','inhaled','other']  
) AS route_classification;  
-- Returns: {"label":"intravenous","score":0.99}
```

Part B — Analysts:

```
SELECT  
    ADMINISTRATIONDIRECTIONS,  
    ADMINROUTE AS stored_route,  
    SNOWFLAKE.CORTEX.AI_CLASSIFY(  
        ADMINISTRATIONDIRECTIONS,  
        ['oral','intravenous','intramuscular',  
        'subcutaneous','topical','inhaled','other']  
    )::VARCHAR AS ai_route  
FROM SI_CHOP.CHOP_SNOW_INTELLIGENCE  
    .V_PHARMACY_ALLMEDICALSCRIPTS  
WHERE ADMINISTRATIONDIRECTIONS IS NOT NULL  
LIMIT 5;  
-- Where stored_route ≠ ai_route = data quality signal
```

Part B — Scientists:

```
SELECT  
    ADMISSION_ID,  
    DISCHARGE_DISPOSITION,  
    SNOWFLAKE.CORTEX.AI_CLASSIFY(  
        'Patient discharged to: ' || DISCHARGE_DISPOSITION,  
        ['high_readmission_risk',  
        'medium_readmission_risk',  
        'low_readmission_risk']  
    )::VARCHAR AS ai_risk_tier,  
    READMITTED_30D AS actual_outcome  
FROM HEALTHCARE_ML.RAW_DATA.ADMISSIONS LIMIT 8;  
-- Discussion: LLM tier vs DISPOSITION_RISK_SCORE feature
```


Exercise 3: AI_FILTER + COMPLETE — Notebook Cell 4 (~7 min)

SE: Both roles run the same cells. No table access needed. Quick but impactful — shows the full pipeline pattern.

AI_FILTER — quality gate (both roles):

```
SELECT
  SNOWFLAKE.CORTEX.AI_FILTER(
    'take as needed',
    'A complete prescription with dose, frequency, and route'
  ) AS passes_gate_1,      -- FALSE → discard
  SNOWFLAKE.CORTEX.AI_FILTER(
    'Administer 500mg amoxicillin orally every 8 hours for 10 days',
    'A complete prescription with dose, frequency, and route'
  ) AS passes_gate_2;      -- TRUE → keep
```

COMPLETE — standardize an informal note (both roles):

```
SELECT SNOWFLAKE.CORTEX.COMPLETE(
  'llama3.1-70b',
  'Standardize to clinical format with dose, route, frequency, duration, monitoring: '
  || '"give 2 tabs twice a day with food for 2 weeks, watch liver"'
) AS standardized_note;
```

Pipeline pattern: AI_FILTER → AI_EXTRACT on passing records only → store structured output.

Quality gating before extraction reduces LLM cost and prevents garbage entering Feature Store / semantic views.

Cortex Code Bridge — Notebook Cell 5 (~5 min, SE-led)

SE: Open your terminal with `coco` (Cortex Code CLI). Type this prompt live — participants watch.

I have these objects in `SI_CHOP.CHOP_SNOW_INTELLIGENCE`:

- `PRESCRIPTION_ORDERS_SV`: semantic view on pharmacy prescriptions
(facts: `script_number`, `costs`, `quantities`; dimensions: `drug`, `route`, `SIG`)
- `MEDICATION_ORDERS_SV`: semantic view on Epic medication orders
- `DRUG_CATALOG_SEARCH`: Cortex Search on drug catalog (`name`, `NDC`, `category`)
- `PRESCRIPTION_DIRECTIONS_SEARCH`: Cortex Search on SIG free text
- `EXTRACT_PRESCRIPTION_ENTITIES(VARCHAR)`: UDF that calls `AI_EXTRACT` on SIG
- `GENERATE_STREAMLIT_APP(VARCHAR)`: procedure for Streamlit dashboards

Generate the SQL to CREATE a Cortex Agent that uses all 6 as tools
and can answer pharmacy questions in natural language.

After `coco` generates the spec:

1. Point out the `tool_spec` array — 4 tool types (`analyst`, `search`, `generic`)
2. Show `EXTRACT_PRESCRIPTION_ENTITIES` in the spec — same UDF from Cell 2
3. Open `sql/infrastructure_setup_guide/06_si_agent_creation.sql` side-by-side
4. Say: "This is exactly what we pre-deployed. Now — let's each deploy our own version."

Deploy Your Own Agent — Personal coco Prompt

SE: Each participant (and you) now runs this second prompt in coco. The username suffix prevents anyone from accidentally overwriting the shared production agent.

Prompt 1 — Generate the spec *(already done on previous slide)*

```
I have these objects in SI_CHOP.CHOP_SNOW_INTELLIGENCE:
- PRESCRIPTION_ORDERS_SV (semantic view)
- MEDICATION_ORDERS_SV (semantic view)
- DRUG_CATALOG_SEARCH (Cortex Search)
- PRESCRIPTION_DIRECTIONS_SEARCH (Cortex Search)
- EXTRACT_PRESCRIPTION_ENTITIES(VARCHAR) (UDF)
- GENERATE_STREAMLIT_APP(VARCHAR) (procedure)
```

Generate the SQL to CREATE a Cortex Agent that uses all 6 as tools for pharmacy questions.

Prompt 2 — Deploy it safely *(new — participants run this)*

Take the CREATE AGENT SQL you just generated. Before running it, make two changes:

1. Rename the agent from:
CHOP_Pharmacy_Intelligence_Agent
to:
CHOP_Pharmacy_Intelligence_Agent_[ASK ME FOR MY SNOWFLAKE USERNAME]
2. Execute the modified SQL in my Snowflake account to create my personal agent.

This avoids overwriting the shared production agent already deployed for the workshop.

Result: Each person ends up with their own agent, e.g.

CHOP_Pharmacy_Intelligence_Agent_JSMITH

visible in **Snowsight** → **Agents**.

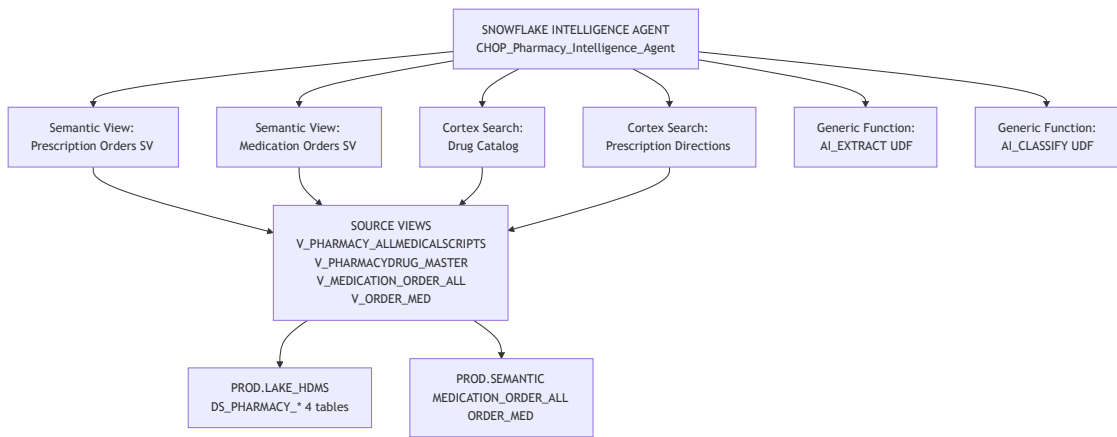
10 minutes



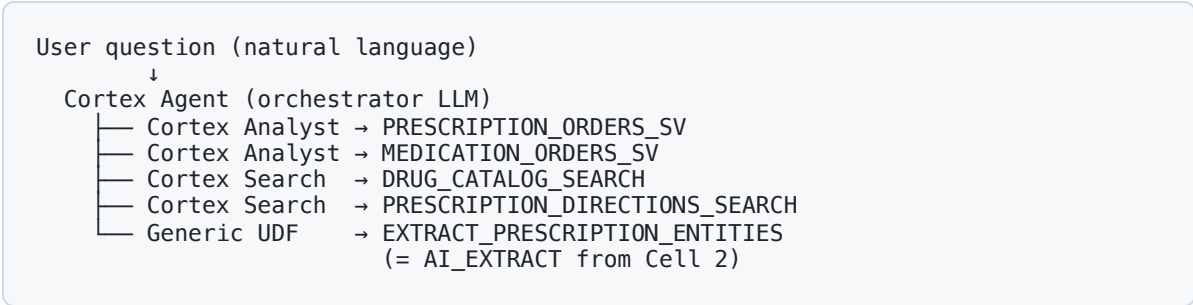
Snowflake Intelligence

Live Agent Demo · ~30 min · Cell 6

SI Architecture — How the Agent Works



Data flow:



Cost controls (already applied):

Object	Control	Cost impact
6 source views	12-month filter + 50K cap	Reduces search index size
Search services	TARGET_LAG= '30 days '	~\$13-15/month (was ~\$150)

SI Tools Inventory — 6 Tools

#	Tool Name	Type	Backing Object	When Agent Uses It
1	Query_Prescription_Orders	cortex_analyst_text_to_sql	PRESCRIPTION_ORDERS_SV	Questions about Rx records, costs, dosing, SIG
2	Query_Medication_Orders	cortex_analyst_text_to_sql	MEDICATION_ORDERS_SV	Epic order questions, therapeutic classes
3	Search_Drug_Catalog	cortex_search	DRUG_CATALOG_SEARCH	Find drugs by name, NDC, or category
4	Search_Prescription_Directions	cortex_search	PRESCRIPTION_DIRECTIONS_SEARCH	Find prescriptions by SIG pattern
5	Extract_Prescription_Entities	generic	EXTRACT_PRESCRIPTION_ENTITIES UDF	Extract structured data from SIG free text
6	Generate_Streamlit_App	generic	GENERATE_STREAMLIT_APP procedure	On-demand Streamlit dashboard generation

SE Note: Roles with USAGE on the agent: CHOP_SNOW_INTELLIGENCE and ML_ENGINEER (grant added in 06_si_agent_creation.sql).
Open agent: **Snowsight** → left nav → Agents → CHOP_Pharmacy_Intelligence_Agent

SI Demo: Agent Call Sheet — Notebook Cell 6

SE: You drive Snowsight. Each analyst picks one question to submit. Show the tool trace after each response.

#	Question (type this in the agent chat)	Tool invoked
1	"What are the top 10 most prescribed drugs by script count?"	Cortex Analyst → PRESCRIPTION_ORDERS_SV
2	"Show me all IV-route prescriptions from the last 30 days"	Cortex Analyst → PRESCRIPTION_ORDERS_SV
3	"Find drugs related to amoxicillin in the drug catalog"	Cortex Search → DRUG_CATALOG_SEARCH
4	"Extract medication name and dose from: 'Give 25mg MTX SQ weekly, hold if WBC < 3'"	Generic UDF → EXTRACT_PRESCRIPTION_ENTITIES
5	"What therapeutic classes have the most medication orders?"	Cortex Analyst → MEDICATION_ORDERS_SV

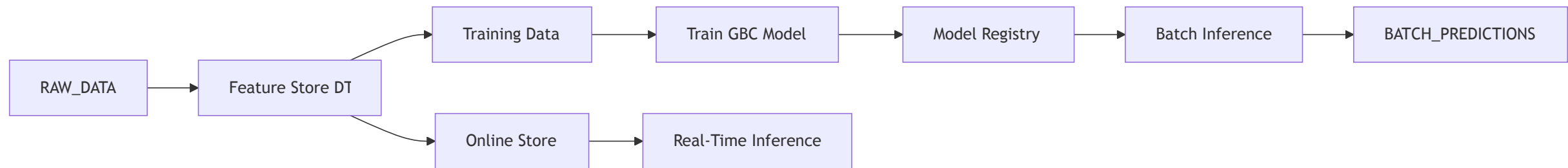
For Scientists — bridge question:

"How would you build a readmission risk agent using HEALTHCARE_ML data? What semantic view would you need? What model would you call as a generic tool?"
Expected answer: semantic view on ADMISSIONS + FEATURE_STORE, generic tool calls READMISSION_PREDICTOR model.

ML Pipeline

Git Integration · Feature Store · Model Registry · ~15 min

ML Architecture — End-to-End Pipeline



Stage	Snowflake Object	Details
Raw Data	HEALTHCARE_ML.RAW_DATA	PATIENTS (5K), ADMISSIONS (10.7K), CLINICAL_MEASUREMENTS
Feature Store	PATIENT_CLINICAL_FEATURES\$V1	Dynamic Table, 33 features, 5-min refresh
Online Store	Same Feature View	1-min lag, sub-second lookup for real-time scoring
Model Registry	READMISSION_PREDICTOR V1	GradientBoostingClassifier, scikit-learn, ROC AUC ~0.85
Batch Inference	BATCH_PREDICTIONS	All patients scored via warehouse <code>mv.run()</code>

Demo notebook: `healthcare-readmission-ml/notebooks/00_demo_walkthrough.ipynb` — 35 cells, 8 sections, full end-to-end in one notebook. Run as `ML_ENGINEER` role.

Git Integration — Code Lives in GitHub

SE: Show the Git repo object in Snowsight. Run `ALTER GIT REPOSITORY ... FETCH` live.

Setup (already done via script 10):

```
-- API integration + secret already created
-- Git repo object:
CREATE OR REPLACE GIT REPOSITORY
  HEALTHCARE_ML.GIT_INTEGRATION.HEALTHCARE_ML_REPO
  ORIGIN = 'https://github.research.chop.edu/
    analytics/healthcare-readmission-ml.git'
  API_INTEGRATION = GITHUB_RESEARCH_CHOP_EDU_API;

-- Sync latest code:
ALTER GIT REPOSITORY
  HEALTHCARE_ML.GIT_INTEGRATION.HEALTHCARE_ML_REPO
  FETCH;

-- Execute from Git:
EXECUTE IMMEDIATE FROM
  @HEALTHCARE_ML.GIT_INTEGRATION.HEALTHCARE_ML_REPO
  /branches/main/production/run_batch_inference.py;
```

Scheduled execution (script 11):

```
-- Task 1: sync code every 60 min
CREATE OR REPLACE TASK
  HEALTHCARE_ML.TASKS.GIT_FETCH_TASK
  WAREHOUSE = HEALTHCARE_ML_WH
  SCHEDULE = '60 MINUTE'
AS
  ALTER GIT REPOSITORY
    HEALTHCARE_ML.GIT_INTEGRATION
    .HEALTHCARE_ML_REPO FETCH;

-- Task 2: score patients after fetch
CREATE OR REPLACE TASK
  HEALTHCARE_ML.TASKS.BATCH_SCORING_TASK
  WAREHOUSE = HEALTHCARE_ML_WH
  AFTER HEALTHCARE_ML.TASKS.GIT_FETCH_TASK
AS
  EXECUTE IMMEDIATE FROM
    @...HEALTHCARE_ML_REPO
    /branches/main/production
    /run_batch_inference.py;
```

Airflow Orchestration

The same ML pipeline · Full DAG visibility · ~10 min demo

Airflow: What It Adds vs Snowflake Tasks

Snowflake Tasks (already running)

- Simple scheduling (cron / AFTER chain)
- SQL + Python via `EXECUTE IMMEDIATE FROM`
- Visible in `INFORMATION_SCHEMA.TASK_HISTORY`
- Runs entirely inside Snowflake

Apache Airflow (Astro CLI, local)

- **Graph view** — visualize the DAG with dependencies
- **Task logs** — per-task stdout with model metrics, row counts
- **XCom** — pass data between tasks (e.g., ROC AUC from train → register)
- **Retries** with backoff per task
- **80+ provider packages** (Slack alerts, dbt, AWS, etc.)

The same 6-stage pipeline, both ways:

Stage	Snowflake Tasks	Airflow DAG
Validate data	<i>(inline)</i>	<code>validate_data_sources</code>
Check features	<i>(inline)</i>	<code>check_feature_tables</code>
Train model	<code>run_training.py</code>	<code>train_model</code>
Register	<code>run_training.py</code>	<code>register_model</code>
Batch score	<code>run_batch_inference.py</code>	<code>run_batch_inference</code>
Validate output	<i>(inline)</i>	<code>validate_predictions</code>

Talking point: Both coexist. Snowflake Tasks = lightweight scheduling. Airflow = production MLOps with observability. Code is **shared** — `src/` modules run unchanged in both paths.

Airflow: Running the Demo

SE: Prerequisites: Docker Desktop running + Astro CLI installed. Do this before the workshop.

Step 1: Configure credentials

Edit `airflow/airflow_settings.yaml` :

```
conn_login: YOUR_SNOWFLAKE_USERNAME
conn_password: YOUR_SNOWFLAKE_PASSWORD
conn_extra:
  account: YOUR_ACCOUNT.us-east-1
```

Step 2: Start Airflow

```
cd airflow/
astro dev start
# First run: 3-5 min (image pull + install)
# Open: http://localhost:8080
# Login: admin / admin
```

Step 3: Trigger

- 1. Find `healthcare_ml_pipeline` in DAG list
- 2. Toggle to unpause
- 3. Click  to trigger a run
- 4. Watch tasks turn **green** sequentially

What to show in the Airflow UI:

View	What to highlight
Graph view	6-task chain with dependency arrows
<code>train_model</code> logs	"ROC AUC: 0.8xxx", training progress
<code>register_model</code> logs	"Registered READMISSION_PREDICTOR V1"
<code>run_batch_inference</code> logs	"Batch predictions saved: N rows"
<code>train_model</code> XCom tab	Return value: model path + ROC AUC

Verify in Snowflake after run:

```
SHOW MODELS IN HEALTHCARE_ML.MODEL_REGISTRY;

SELECT COUNT(*) AS scored_patients
FROM HEALTHCARE_ML.INFERENCE.BATCH_PREDICTIONS;
```

Stop Airflow:

```
astro dev stop
```

Summary

What Participants Leave With

AISQL skills:

- Extract structured entities from free text (`AI_EXTRACT`)
- Classify and guardrail data quality (`AI_CLASSIFY` , `AI_FILTER`)
- Standardize and transform text (`COMPLETE`)
- All work inline in any `SELECT` statement

Snowflake Intelligence:

- Natural language → SQL via semantic views
- Agent orchestrates 6 tools automatically
- Cost-controlled (~\$13-15/month search)

ML Pipeline:

- Feature Store with 33 engineered features
- Model Registry with versioned GBC model
- Git integration: code lives in GitHub, runs in Snowflake
- Scheduled Tasks: auto-fetch + auto-score daily

Airflow:

- Same pipeline with full DAG visibility
- Per-task logs, XCom data passing, retries
- `astro dev start` → trigger → watch green

Files in Git repo (`sfc-gh-moahmed/Workshops`):

`sql/CHOP_participant_readiness_check.sql` • `healthcare-readmission-ml/notebooks/workshop_aisql_si_exercises.ipynb` • `docs/CHOP_workshop_plan.html` • `docs/CHOP_workshop_plan.pdf`